MATH 243: Discussion Worksheet 9

Topics: 14.8 Lagrange Multipliers; 15.1 Double Integrals over Rectangles
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1. Use the method of Lagrange multipliers to find the maximum and minimum values of f subject to the given constraint.
   (a) $f(x, y) = 9xe^y$, subject to $x^2 + y^2 = 2$.
   (b) $f(x, y) = 81x^2 + y^2$, subject to $4x^2 + y^2 = 9$.

2. A rectangle in $\mathbb{R}^2$ has one vertex at the origin, two sides along the positive coordinate axes, and its opposite vertex in the first quadrant on the curve $x^3 + y^3 = 16$. Use the method of Lagrange multipliers to find the dimensions of the rectangle with the largest area.

3. Find the absolute maximum and absolute minimum values of the function
$$f(x, y) = 2x^2 - y^2 + 6y$$
on the disk $x^2 + y^2 \leq 16$.

4. Evaluate each double integral.
   (a) $\displaystyle\iint_R \left(y + \frac{x}{y^2}\right) dA$, where $R = \{(x, y) \mid 0 \leq x \leq 2,\ 1 \leq y \leq 2\}$.
   (b) $\displaystyle\iint_R x\cos(xy)\, dA$, where $R = [0, \pi] \times [0, 1]$.
   (c) $\displaystyle\iint_R \frac{1}{(2x + 3y)^2}\, dA$, where R is the rectangle with vertices $(0, 1),\ (1, 1),\ (1, 2),\ (0, 2)$.
   (d) $\displaystyle\iint_R \frac{y}{(1 + xy)^2}\, dA$, where $R = \{(x, y) \mid 0 \leq x \leq 2,\ 0 \leq y \leq 1\}$.

5. Find the volume of the solid that lies under the hyperbolic paraboloid
$$z = 3y^2 - x^2 + 2$$
and above the rectangle $R = [-1, 1] \times [1, 2]$.

6. Find the average value of $f(x, y) = e^{2x-y}$ on the rectangle $R = [0, \ln 3] \times [0, \ln 2]$.

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Method of Lagrange Multipliers. To find the maximum and minimum values of $f(x, y)$ subject to the constraint $g(x, y) = k$ (assuming that such extrema exist, that f and g are differentiable, and that $\nabla g \neq \mathbf{0}$ on the constraint curve), we find all values x, y, λ such that
$$\nabla f(x, y) = \lambda \nabla g(x, y) \qquad \text{and} \qquad g(x, y) = k.$$
Then evaluate f at each corresponding point (x, y). The largest of these values is the maximum value, and the smallest is the minimum value.

Double Integrals over Rectangular Regions.
If f is continuous on the rectangle $R = [a, b] \times [c, d]$, then

$$\iint_R f(x, y)\, dA = \int_a^b \int_c^d f(x, y)\, dy\, dx = \int_c^d \int_a^b f(x, y)\, dx\, dy.$$

If $f(x, y) \geq 0$ on R, then $\iint_R f(x, y)\, dA$ represents the volume of the solid that lies above R and below the surface $z = f(x, y)$.

The Average Value of a Function of Two Variables.
The average value of a function f over a region R of area $A(R)$ is

$$f_{\text{avg}} = \frac{1}{A(R)} \iint_R f(x, y)\, dA,$$

provided that $0 < A(R) < \infty$.

Suggested Textbook Problems
Section 14.8: 1-29
Section 15.1: 1-49, 53-56